



Animesh Choudhury

PhD Researcher in Earth & Atmospheric Sciences | ML for Weather Forecasting | Remote Sensing Specialist



PROFESSIONAL SUMMARY

PhD researcher in Earth and Atmospheric Sciences with experience in geospatial analysis, remote sensing, and data-driven approaches to weather and climate studies. My work focuses on exploring the application of machine learning and AI in improving weather forecasting and understanding climate variability. I have had the opportunity to work across diverse domains including atmospheric modeling, drought monitoring, air quality analysis, and satellite data interpretation. I have presented my work at conferences and contributed to peer-reviewed publications. I am continuously looking to learn, improve, and contribute to the growing intersection of geospatial science and AI for addressing real-world environmental challenges.

EDUCATION

PhD in Earth and Atmospheric Sciences | 2022 - Current

National Institute of Technology, Rourkela, India

Post Graduate Diploma in Remote Sensing & GIS | 2017-2018

Indian Institute of Remote Sensing (IIRS), Dehradun, India

M.Sc in Environmental Science | 2015-2017

Asutosh College, Calcutta University, Kolkata, India

B.Sc in Physics | 2009-2013

Raghubanpur College, Burdwan University, West Bengal, India

WORK EXPERIENCE

Junior Research Fellow (PhD Researcher)

National Institute of Technology, Rourkela, India | 15/02/2022 - 30/06/2025

- Developing data-driven weather forecasting systems using Pangu-Weather architecture and IMDAA reanalysis data
- Creating BharatBench dataset for ML-based weather forecasting over India
- Conducting high-resolution mesoscale modeling of urban thunderstorms using WRF
- Analyzing land-atmosphere interactions and urban heat island effects

Junior Research Fellow

Department of Remote Sensing and GIS, Vidyasagar University, Midnapore, India | 21/01/2020 - 07/02/2022

- Long-term monitoring of droughts using satellite-based indices (SPEI, VCI, TCI)
- Developed ML models for prediction of agricultural droughts
- Analyzed regional variation of drought parameters across India
- Published research in Journal of Environmental Management

Junior Research Fellow

DST-Mahamana Centre of Excellence in Climate Change Research, Banaras Hindu University, Varanasi, India | 13/06/2019 - 19/01/2020

- Observation of horizontal and vertical distribution of air pollutants using CALIPSO LIDAR
- Analyzed aerosol transport patterns over Indo-Gangetic plain
- Integrated ground-based and satellite observations for air quality assessment

Junior Research Fellow

Indian Institute of Remote Sensing, Dehradun, India | 21/10/2018 - 30/03/2019

- Snow cover area mapping of Northwest Himalaya using MODIS, Sentinel, Landsat datasets with Google Earth Engine
- Data collection, installation, and supervision of automatic weather station (AWS), snow precipitation gauge (SPG), and snow pack analyzer (SPA)
- Published research in Remote Sensing journal

Junior Research Fellow

CSIR - Institute of Himalayan Bioresource Technology, Palampur, India | 12/08/2018 - 14/10/2018

- Collection of spectral signatures using ASD Fieldspec spectroradiometer
- Correlated spectral data with bio-physical parameters
- Classification of tree species using satellite-based hyperspectral data

SELECTED PUBLICATIONS

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- [1] Choudhury, A., Panda, J., & Bhattacharyya, S.. **A Modeling Effort for Medium-Range Weather Prediction over India with a Modified Pangu-Weather Architecture**. *Asia-Pacific Journal of Atmospheric Sciences*, 2026.
 - [2] Choudhury, A., Panda, J., & Mukherjee, A.. **Urban growth analysis and prediction over six Indian cities and its association with pre-monsoon LST and rainfall**. *Journal of Earth System Science*, 2026.
 - [3] Panda, J., Mukherjee, A., Sarkar, A., Kumar, G., & Choudhury, A.. **Analysis and Prediction of PM2.5 Concentrations Over Indian Megacities Using Geospatial Techniques and Deep Learning Models**. *Geospatial Technologies*, 2026.
 - [4] Roy, D., Panda, J., Mukherjee, A., Choudhury, A., & Panda, S. K.. **A long-term analysis of pre-monsoon thunderstorm characteristic features over major urban areas of India**. *Acta Geophysica*, 2026.
 - [5] Choudhury, A., & Panda, J.. **Development of a Data-driven weather forecasting system over India with Pangu-Weather architecture and IMDAA reanalysis Data**. *In Preparation*, 2025.
 - [6] Choudhury, A., Panda, J., & Mukherjee, A.. **BharatBench: Dataset for data-driven weather forecasting over India**. *In Preparation*, 2024.
 - [7] Mukherjee, A., Panda, J., Choudhury, A., Singh, S., & Bhattacharyya, S.. **Analyzing urban footprints over four coastal cities of India and the association with rainfall and temperature using deep learning models**. *Urban Climate*, 2024.
 - [8] Singh, S., Mukherjee, A., Panda, J. et al.. **Analysis and Forecasting of Temporal Rainfall Variability Over Hundred Indian Cities Using Deep Learning Approaches**. *Earth Systems and Environment*, 2024.
 - [9] Choudhury, A.. **Drought trend and its association with land surface temperature (LST) over homogeneous drought regions of India (2001–2019)**. *Discover Water*, 2024.
 - [10] Panda, J., Mukherjee, A., Choudhury, A., Biswas, S.. **Urban Heat: UHI and Heat Stress Threat to Megacities**. *Climate Crisis: Adaptive Approaches and Sustainability. Sustainable Development Goals Series. Springer, Cham*, 2023.
 - [11] Budakoti, S., Singh, C. & Choudhury, A.. **Transport of a severe dust storm from Middle East to Indian region and its impact on surrounding environment**. *International Journal of Environmental Science and Technology*, 2022.
 - [12] Choudhury, A., Dutta, D., Bera, D., Kundu, A.. **Regional variation of drought parameters and long-term trends over India using standardized precipitation evapotranspiration index**. *Journal of Environmental Management*, 2021.
 - [13] Choudhury A, Yadav AC, Bonafoni S. **A Response of Snow Cover to the Climate in the Northwest Himalaya (NWH) Using Satellite Products**. *Remote Sensing*, 2021.

CONFERENCE PAPERS & PROCEEDINGS

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- [1] **A Step Towards Building Medium-Range Data-Driven Weather Forecasting System Over India**. *IEEE Conference*, 2024.
 - [2] **Monitoring of Surface Water of East Kolkata Wetlands (1984-2015) with Global Surface Water Dataset**. *ACRS 2020*, 2020.
 - [3] **Role of Earth Observation Data and Hydrological Modeling in Supporting UN SDGs in North West Himalaya**. *ISPRS Archives*, 2020.
 - [4] **Simulation of an Extreme Dust Episode using WRF-CHEM**. *IASTA 2018*, 2018.

TECHNICAL SKILLS

Weather & Climate Modeling: WRF-CHEM, Pangu-Weather, Mesoscale Modeling, IMDAA Reanalysis, Climatological Data Analysis

Machine Learning & AI: Deep Learning, TensorFlow, PyTorch, Time Series Forecasting, Neural Networks, Data-Driven Models

Remote Sensing: MODIS, Sentinel, Landsat, SNAP, ENVI, Idrisi, Hyperspectral Analysis, ASD Fieldspec

GIS & Geospatial: Google Earth Engine, ArcGIS, QGIS, Spatial Analysis, Geospatial Modeling

Programming & Data Science: Python, R, SQL, Data Processing, Data Analysis, Data Visualization

Climate Analysis: Drought Monitoring, SPEI/VCI/TCI Indices, Climate Trends, Snow Cover Mapping, Atmospheric Science

POSTER, ONLINE TRAINING, WEBINAR

- **Poster Presentation at AOGS 2025** on "Regional Data-driven Weather Forecasting Over India with Pangu-weather Architecture and IMDAA Reanalysis" in the 22nd Annual Meeting of Asia Oceania Geosciences Society held at Sands Expo & Convention Centre Marina Bay Sands, Singapore during 27 Jul - 01 Aug, 2025
- **National Symposium Participation** in the Tropment-2024, Indian Meteorological Society held at NIT Rourkela during 10-12 December 2024
- **Organizing Member and Speaker** in the National Workshop on Drought Monitoring and Early Warning using Geospatial Techniques held at Department of Remote Sensing and GIS, Vidyasagar University during 24th February, 2021
- **Poster Presentation at AAPW-5** on "Aerosol vertical distribution over lower Gangetic plain using CALIPSO LIDAR" in the Asian Air Pollution Workshop during 5th-7th November, 2019
- Completed the "Machine Learning for Earth Systems Modelling — Course 2: Architectures, Data, and Prediction (23 hours)" organised by European Centre for Medium-Range Weather Forecasts (ECMWF), under the Destination Earth Initiative, in collaboration with Karlsruhe Institute of Technology (KIT) and Wageningen University & Research during 27 July 2026
- Completed the "Recent Trends in Time Series Forecasting using Deep Learning Models" organised by Department of Computer Science & Engineering, NIT Rourkela during December 25-29, 2023
- Completed the "Deep Learning Application for Smart Cities" organised by Department of Computer Science & Engineering, NIT Rourkela during September 11-15, 2023
- Completed the "Machine Learning in Weather & Climate (MOOC)" organised by ECMWF in partnership with IFAB during January to April, 2023
- Completed the "Capacity Building - Training School on WRF Modeling System" organised by C-DAC and SAMA under National Supercomputing Mission (NSM) during 3rd to 20th April 2023
- Completed the "Open Classroom Online Lecture Series in Atmospheric and Climate Sciences" organised by SAMA and Centre for Atmospheric Sciences, SRM Institute during 7-24 April 2023
- Completed the "Geospatial Modeling Driven Urban Hazard and Risk Analysis" organised by IIRS, Dehradun during 13-02-2023 to 17-02-2023
- Completed the "Machine Learning (ML) and Deep Learning (DL) for Earth Observation (EO) Data and Application" organised by DA-IITC in Collaboration with SAC, ISRO during 9th-13th August, 2021
- Attended a webinar on "Thunderstorm and Lightning Physics" organized by Department of Physics (DST-FIST & DST-PURSE Sponsored), Koneru Lakshmaiah Education Foundation (KLEF), in association with the Indian Aerosol Science & Technology Association (IASTA) on 24 April 2026
- Attended a webinar on "Environment and Ecosystem Restoration" organized by Department of Physics, K.L.E.F., Vijaywada, Andhra Pradesh on 5th June, 2021
- Attended a webinar on "Geospatial Techniques in drought characterization and impact assessment" organized by Albedo Foundation on 26th September, 2020

Last updated: September 2026